Week 14 Assignment

Nonparametric alternatives and choosing the right test

This is the last weekly assignment before final project presentations — nice work getting through the whole sequence. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 15's presentations. See the syllabus for the late policy.

Part 1: Applied

Using the built-in `chickwts` dataset:

1. Run a Kruskal-Wallis test of `weight` by `feed`.
2. Run the parametric one-way ANOVA equivalent for comparison.
3. Do the two tests agree on whether there's a significant difference among feed types? If they disagreed, which would you trust more, and why?

Part 2: Choosing a test, cover to cover

For each scenario, name the test or model from this course (any week) that you'd reach for, and briefly justify your choice:

1. Comparing mean growth between 2 treatment groups, with no reason to doubt normality or equal variance.
2. Comparing mean growth between 2 treatment groups, where one group has a couple of extreme outliers you can't justify removing.
3. Testing whether a continuous covariate predicts a binary outcome.
4. Testing a treatment effect when measurements are repeated across multiple sites, with random variation in both the baseline and the treatment effect from site to site.